Answers to the expert input request

Renewable energy assessment app: reference values and rules for a worldwide tool (version 2)

Prepared in reply to Timi's 12 questions. Research date: 10 November 2026. This version covers Europe (including Central and Eastern Europe), North America, Latin America and Africa. Countries are used as examples inside climate zones, not as special cases. All sources are free-to-read web pages, reports or open-access papers (see the source list).

Tag	Meaning
[Sourced]	Number or rule comes directly from a published source listed at the end (marked [S1], [S2]…).
[Derived]	Our own calculation from sourced inputs. The working is shown so it can be checked.
[Judgement]	Engineering judgement or a reasonable default where no clean source was found. Please have an expert confirm these before they go into the scoring.

Climate zones used throughout. Zone boundaries should be set from heating and cooling degree-days of the user's location [Judgement].

Zone	Examples	Main energy needs and design issues
Cold	Nordic countries, Canada, northern USA, Russia, high Andes and Patagonia	Heating dominates; insulation and heat pumps matter most; snow on solar panels.
Cool-temperate	Central and Eastern Europe (e.g. Poland, Romania, Hungary), Germany, UK, northern USA	Heating is the largest use; winter has little sun; good fit for heat pumps and insulation.
Mild / Mediterranean / subtropical	Southern Europe, California, central Chile, Pampas, Uruguay, southern Brazil, Cape Town, North African coast	Small heating need plus cooling; strong solar; hot water often solar.
Hot-humid / tropical	Northern and coastal Brazil, Central America, Caribbean, lowland West, Central and East Africa	Cooling and cooking; little or no heating; frequent outages in many countries; fuel-based cooking common.
Hot-arid	Atacama, northern Mexico, Sahel, Sahara, Namibia, Egypt	Very strong sun; dust (soiling) on panels; cooling load; water scarcity.

Tropical and subtropical **highlands** (Andes, Mexico City and Bogotá, Ethiopian and Kenyan highlands, South African Highveld) have cool nights and some heating even near the equator [Judgement].

1. The building (year built, renovation, insulation, windows)

1a. Heating demand by year built and insulation

Reference for a single-family house in a **cool-temperate** climate (German-type benchmarks, measured and weather-corrected), in kWh of heat per m² of heated floor area per year:

Building class	Heating demand (kWh/m²·yr)	Notes
Built before 1977, no renovation	Above 145; typically 150–250	Techem measured classes: over 145 [S2]. Energy label scale runs from 30 (A+) to over 250 (H) [S2]. Allow a "very poor" class for old adobe or uninsulated houses; Central and Eastern European studies report figures around 400 for the worst types [S5].
Built 1977–2002 (first insulation rules)	90–145	Techem class for houses under the 1977/1995 rules [S2].
Built after 2002 (modern codes)	Under 90; new builds 30–75	Techem class [S2]; new-build range [S2].

Building class	Heating demand (kWh/m²·yr)	Notes
Passive-house level	About 15	Passive House standard [Judgement: well-known figure, not re-checked].

Other zones. Insulation level matters as much as climate. In a TABULA comparison, a German house from 2001–2016 needed 76.8 kWh/m² for heating in a temperate climate, against 31.8 for a Spanish house of the same period, even though the Spanish walls were less insulated [S1]. So the mild-zone value is about 0.4 of the cool-temperate value [Derived: 31.8 ÷ 76.8]. For cold zones, scale by the heating degree-day ratio (often 1.3–1.8 against central Europe) but note that Nordic houses are far better insulated, so absolute values are often similar [Judgement]. In **hot-humid and hot-arid zones** space heating is close to zero; the building question becomes cooling (roof colour and insulation, shading, window type), and the app should use cooling degree-days instead [Judgement]. TABULA covers European countries only; for the Americas and Africa use national building-energy labels where they exist, and otherwise a simple user-declared “poor / average / good” insulation class [Judgement; national schemes not checked].

Wall and window U-values (W/m²K) differ strongly by country [S1]:

Country	Wall U, 1970–85	Wall U, 2001–16	Window U, 1970–85	Window U, 2001–16
Germany	1.00	0.30	2.80	1.40
United Kingdom	1.60	0.35	4.80	1.85
Spain	1.33	0.48	4.59	3.09
Sweden	0.21	0.20	2.01	1.87

Central and Eastern European stock is old and often uninsulated: about half of Hungarian homes lack insulation and heating takes about 72–74% of household energy there [S5]. Hungary has a TABULA/EPISCOPE typology [S5]; we did not confirm sets for other countries, so please check the TABULA WebTool before fixing national tables.

1b. How much a renovation cuts heating demand, and how to count an old one

- **Size of the effect.** A German single-family house from 1958–68 is calculated at 211 kWh/m² unrenovated, 97 after a standard renovation package and 52 after an advanced one [S4]. That is about −54% and −75% [Derived].
- **Calculated vs real savings.** German experience shows calculated savings of about 33% but measured savings of about 25%, because people heat differently from the model [S3]. Apply a realism factor of about 0.75 to calculated savings [Derived: 25 ÷ 33].
- **Per element** (rough shares of heat loss in a poorly insulated house: roof 15–25%, walls 20–30%, windows 25–40%, floor 5–15%, ventilation 15–25%) [Judgement; the window share is supported by [S6]].
- **Renovation 10 or 20 years ago.** Do not discount for age: mineral wool and EPS keep most of their performance over 20 years [Judgement]. Instead, count the renovation at the **standard of its year**. A wall insulated 20 years ago usually has 5–8 cm (U about 0.5–0.6) and one done 10 years ago 10–14 cm (U about 0.25–0.35) [Judgement]. If the user does not know the thickness, assume the thinner value for work older than 15 years.

Suggested rule for the app

Ask, per element (roof, walls, windows, floor): “not done / done before 2005 / done 2005–2015 / done after 2015”, and set U-values from the national typology for that period. Then show the result as a range, because renovation quality varies.

1c. How much window type matters

Window type	U-value, whole window (W/m²K)	Heat loss vs single glazing
Single glazing	4.5–5.8 (use 4.8)	100%
Older double glazing	2.5–3.3 (use 2.8)	about 55% (−45%)

Window type	U-value, whole window (W/m²K)	Heat loss vs single glazing
Newer double, low-e coating, gas fill	1.1–1.6 (use 1.3)	about 25% (−75%)
Triple glazing	0.6–1.0 (use 0.8); older triple up to 1.8	about 15% (−85%)

U-values from [S6]; percentages [Derived: U ÷ 5.0]. In a poorly insulated home, windows can account for 25–40% of heat loss [S6], so replacing single glazing with modern double glazing cuts whole-house heating demand by roughly 20–30% [Derived: 25–40% × 75%]. In hot zones the same logic applies to cooling: window shading and low solar-gain glass matter more than the U-value [Judgement].

2. Heating, hot water and cooking

Hot water per person

Case	Energy per person per year	Source / note
Standard assumption (about 40 L/day at 60 °C)	800 kWh	Used as a flat figure in German district-heating rules [S2]; consistent with 40 L/day per person in the Swiss SIA 2024 standard [S7].
Careful use (20–25 L/day)	440–550 kWh	[S7]
Heavy use (50–70 L/day, baths)	up to 1,600 kWh	[S7]
Sweden, family of four (incl. tank losses)	875–1,625 kWh (3,500–6,500 per household)	[S7]
USA (water heating about 3,000 kWh per household)	about 1,200 kWh	[Derived: 3,000 ÷ 2.5 persons; 2008 US average quoted in [S9]]
Brazil (electric shower in about 73–85% of homes)	The shower is about 19–26% of residential electricity	[S35]. A solar heater with electric backup cut water-heating electricity by about 70% and total home electricity by about 36% in one study [S35].

Add 400 kWh/yr for a well-insulated storage tank, or up to 1,200 kWh/yr for an old one [S7]. Where cold-water inlet is warmer (15 °C vs 10 °C), reduce by about 10% [Derived: 45 ÷ 50].

Cooking per person

In Europe, cooking is about 6.4% of household energy use (2024) [S8]. New appliances use far less than the average: an electric hob about 231 kWh and an oven about 84 kWh per household per year; a gas hob about 290 kWh and an oven 147 kWh of gas [S8]. The US figure is about 1,000 kWh per household [S9].

In **Sub-Saharan Africa**, national surveys (2012–2017) show wood as the main cooking fuel for about 63% of people, charcoal 15%, LPG 7% and electricity 7% [S34]. A benchmark of about 1 GJ (about 280 kWh) of final cooking energy per person per year is used in electric-cooking studies [S34], and earlier work assumed 308–397 kWh of electricity per person per year, which efficient pressure cookers and slow cookers can cut a lot [S34]. A clean-cooking study assumed 25 kg of LPG per person per year, about 340 kWh [S34; Derived: × 13.6 kWh/kg]. With an improved stove, firewood use can fall to about 0.85 kg per person per day [S34].

Suggested rule for the app

Default cooking: **150 kWh per person** (electric) or **200 kWh per person** (gas), range 100–350 in Europe [Derived: appliance data above ÷ 2.1–2.3 persons, rounded up for real use]. Use about 400 kWh per person for the USA [Derived].

Africa and parts of Latin America: use **300–350 kWh per person of final energy** for LPG or electric cooking, and treat wood or charcoal as a user-declared fuel [S34; Derived].

If the user does not know their bills

Item	Default	Source
Electricity per household per year (all uses)	EU average 3,600 kWh (Central and Eastern Europe often lower, e.g. Romania 1,800); Spain 4,169; UK 3,658; USA about 10,400 (EIA 2026) to 11,200; Canada 11,305; Australia 5,919; South Africa 2,614; China 2,180; India 1,005	[S9][S29]
Brazil and Mexico	About 500 kWh per person, so roughly 1,500–2,000 kWh per household	[S29; Derived]
Lighting and appliances per m² (kWh/yr), Europe	Germany 25, France 30, Spain 24, Italy 14.5, UK 38, Sweden 58	[S1] Table 3
Average dwelling size, Europe	UK 75 m², Germany 84, Sweden 91, France 92, Spain 94, Italy 96	[S1] Table 4
Hot water	800 kWh per person in Europe (see above)	[S2]
Heating	Typology table (section 1) × floor area × degree-day ratio; none in hot zones	[S1][S2]

3. Electricity and gas from the last 3 months

Is it sound to estimate yearly use from 3 months, adjusted for the season?

Yes, but only with a degree-day method, and with a stated uncertainty. The standard approach (PRISM) fits energy use against degree-days to separate a weather-independent base load from heating (or cooling), then scales to a normal year. It is the usual way to weather-correct bills [S10]. Accuracy depends on how much data is used: short periods are weaker because other factors (solar gains, behaviour, hot water) matter more in months with few degree-days [S10; Judgement on the size of the error].

- **Best case:** the 3 months include at least one month with no heating or cooling, so base load can be read directly.
- **Worst case:** 3 mid-winter (or mid-summer in hot zones) months. Base load and weather-driven use cannot be separated, and the estimate may be off by 20–30% [Judgement].
- **Electricity without electric heating, air conditioning or EV:** seasonality is small (about ±10–15%), so a simple ×4 with a seasonal factor is acceptable [Judgement]. **With air conditioning** (hot zones) the seasonal swing is large, so use cooling degree-days.

Suggested rule for the app

Ask for **the month names** with each bill. Compute: base load from the lowest-use month; weather-driven use = (bill − base) × (annual degree-days ÷ degree-days of those months), from the same weather station.

Show the result as a range (e.g. ±20%). Cross-check against the typology: if the bill-based figure is more than 1.5× or less than 0.5× the typology value, ask the user to confirm area and fuel.

Offer an “enter 12 months” option for better accuracy.

When to warn “very high for a home”

Average household use varies about 100-fold between countries [S29]. A fixed number cannot work worldwide, so compare with the user’s own country. Heat pumps add about 170–420 kWh/month and an EV about 250 [Derived from heat pump 2,000–5,000 kWh/yr and EV about 3,000 kWh/yr, [S25]].

Country or region (example)	Average per month	Warn “check” at 2×	Warn “very high” at 3×
India	84	170	250
Brazil / Mexico	about 125–165	about 300	about 450
South Africa	218	440	650

Country or region (example)	Average per month	Warn “check” at 2×	Warn “very high” at 3×
Central and Eastern Europe (e.g. Romania)	150	300	450
EU average	300	600	900
Spain	347	700	1,040
Australia	493	990	1,480
USA (EIA 2026)	864	1,730	2,590
Canada	942	1,880	2,830

Suggested rule for the app
Use **3× the user’s national average** for “very high” and 2× for “check”, **added to** declared heat pump (+400), EV (+250) and, in hot zones, air conditioning (+200–500 kWh/month per split unit in heavy use [Judgement]). Values in kWh per month [Judgement based on the averages in [S9][S29]]. For gas or LPG, warn when use exceeds twice the typology expectation for the home.

4. Power cuts

When is a battery worth recommending?

How often and how long the grid fails differs by a factor of hundreds:

Region	Supply quality	Source
Western Europe (e.g. Germany)	About 12 minutes of unplanned outage per customer per year (2024)	[S11]
Central and Eastern Europe (e.g. Romania)	229 minutes in 2024; about four hours on average in 2025	[S11]
All of Europe	7 to 2,371 minutes per year in 2024	[S11]
Latin America and Caribbean	About 12 interruptions a year on average in 2015, far below Africa and South Asia; storms and hurricanes can cause multi-day events	[S30]; storms [Judgement]
South Africa	6–9 outages a month in 2025; 73–132 hours a month without grid power for households with no solar (data from solar-battery customers, so it may lean towards problem areas)	[S30]
Sub-Saharan Africa, average	About 741 hours and 253 interruptions per year in 2015 (pulled up by a few extreme countries)	[S30]

Situation	Recommendation
Under about 60 min/yr, no critical load	Do not recommend a battery for backup. Mention only if it pays through self-consumption.
60–300 min/yr, or 1–3 short cuts a month	Optional backup add-on: **5–10 kWh** with a 3–5 kW inverter.
Over 300 min/yr, cuts at least monthly, or medical equipment, well pump, remote work	Recommend **10–13 kWh class** with backup (islanding) capability.
Frequent or daily cuts (over about 20 hours a month, or weekly multi-hour cuts)	A **hybrid solar-plus-battery** system becomes the main recommendation; size for the evening and night: **10–15 kWh** for a light household, 15–25 kWh with air conditioning [S33; Judgement]. Compare with the fuel cost of a generator.
Long events (storm, wildfire shut-offs of 24–72 h)	Battery plus solar, or a generator; critical loads 10 kWh/day need about 30 kWh for 3 days without sun [S12].

Thresholds [Judgement]. Important: a normal grid-tied solar system **stops working during a grid outage**, so PV alone gives no backup. A battery-and-inverter system must be built for backup [S12; Judgement on the grid-tie point].

How big?

- Essential loads (fridge, lights, router, phone chargers, heating circulation pump or fans) typically use **5–10 kWh/day** [S12].
- A guide for Nigeria sizes a light household (12–18 kWh/day) at 4–6 kWp plus a 10–15 kWh battery and a 4–5 kW inverter, and a household with air conditioning (20–30 kWh/day) at 6–10 kWp plus 15–25 kWh [S33].
- Formula: battery kWh = critical kWh/day × days of autonomy ÷ (usable depth of discharge about 0.9 × efficiency about 0.9) [S12; Derived].
- For cuts “a few times a month”, each lasting hours, **5–10 kWh** is enough. A 10 kWh battery runs essential loads for about 10–24 hours [S12].
- Motors (well pumps, compressors, air conditioners) start with 4–7× their running current, so the inverter rating, not the battery size, may be the limit [S12].

Incentives for batteries differ and change fast; for example, a residential battery grant in Hungary requires at least 10 kWh [S20], and Romania adopted a storage requirement for larger prosumer systems in 2024 [S19]. The app should read these from a country table, not hard-code them.

5. Free space and its size

Option	Space needed	Notes and source
Solar panels on roof	**4.3–4.8 m² per kWp** (430 W modules), up to 5 m² per kWp with older modules	5 kWp ≈ 20–25 m²; 10 kWp ≈ 40–50 m² [S13].
Solar panels on ground	**10–20 m² per kWp**	Panel area divided by a ground-cover ratio of 0.4–0.5 [Derived]; rows need more spacing at higher latitudes (5.8–8.2 m pitch in northern Europe on flat roofs) [S14].
Solar water heater	**1–1.5 m² flat-plate per person** (0.7–1 m² evacuated tube); 4 people: 4–6 m²	Tank 300–500 L (about 80 L per person) [S16]. Yield about 350 kWh/m² (flat plate) to 450 (tube) per year in central Europe; about 60% of hot water covered [S16]. Sunnier zones yield more [Judgement].
Air-source heat pump	Outdoor unit outside only (never indoors) with free airflow; indoor cylinder or buffer needed	Typical footprint about 1–2 m² with clearances [Judgement: follow manufacturer minimum distances]. [S15] Only relevant where there is a heating season.
Ground-source heat pump, horizontal	About **2.5× the house footprint**; roughly 200–250 m² for 8 kW; up to 50 m² per kW in damp clay soil	[S15]
Ground-source heat pump, boreholes	**10–20 m of borehole per kW** (8 kW: 80–160 m); 5–10 m between holes; hole about 150 mm wide	[S15]
Small wind turbine	At least about 1 acre (4,000 m²) of open land; tower often 18–30 m	Setback about 1.5× tower height; blade tips 10 m above anything within 150 m [S17].

Is “a medium roof is about 60 m²” realistic?

Yes as the area of **one roof plane** of a 100–120 m² house, but the **usable** area is lower: after edge setbacks (about 0.4–0.5 m), chimneys, roof windows, water tanks and shade, about 65–80% is usable [Judgement]. So 60 m² gives about 40–50 m², or **8–10 kWp** [Derived: 42 m² ÷ 4.6 m²/kWp ≈ 9]. This is a large home system; typical European homes install 5–8 kWp, up to 15 kWp with a heat pump and EV [S25]. In Latin America and Africa many homes have **flat concrete roofs** shared with water tanks, stairwells and laundry space, so ask for “free flat roof area” rather than roof planes [Judgement].

Output lost for roof orientation (northern hemisphere, mid-latitudes)

Roof	Annual output vs best (south, 30–35°)	Notes
South-east or south-west	90–97%	[S14]
East or west, pitched roof	75–85% (UK/Central Europe); 85–92% at lower tilt	Loss is smaller at lower latitudes: about 8–9% at 37–42°N, 12–13% at 50–60°N [S14].
East-west pair on a flat roof (10–15° tilt)	about 85–90%	Madrid: −10%; London flat E–W: −15%, pitched E–W: −25% [S14]. Fits 15–25% more panels per m² [S14].
Flat roof, panels flat (0°)	about 85–88%	Use tilt frames where possible [S14].
North-east or north-west	55–65%	[S14]
North	40–55%	Not recommended unless prices are very high [S14].

Suggested rule for the app

In the **southern hemisphere** (Chile, Argentina, Brazil, southern Africa, Australia) swap north and south: the best roof faces north. Near the equator (below about 15° latitude) orientation matters little and a flat roof is close to optimal. Show output loss as a number and keep flat-roof E–W as an option, because it often wins on self-consumption [S14].

Dust and heat in hot zones

- **Soiling (dust):** typical annual losses are 1–7% in many places [S36]; about 3–5% on average worldwide [S36]. North Africa, the Middle East and the northern Atacama are higher [S36]; Santiago, Chile loses about 7% from city pollution [S36]. In the hyper-arid Atacama, hard “cemented” dust can cost up to about 9.8% a year [S36]. Rain-free regions need regular rinsing (see section 7).
- **Heat:** hot panels produce less; allow roughly 8–12% extra loss in hot zones compared with a 25 °C reference [Judgement]. Batteries and inverters also age faster in heat, so keep them shaded and ventilated [Judgement].
- **Sun available:** Brazil’s average daily irradiation on a horizontal plane ranges from about 5.2 kWh/m² (south) to 5.9 (northeast) [S35]; the Atacama exceeds 8 kWh/m² a day in places [S36].

6. Ownership (renting, flats)

Plug-in (“balcony”) solar is the main option for tenants in some countries, but its legal status varies:

Region	Status (November 2026)	Source
Germany	Legal up to 800 W inverter output (modules up to 2,000 W); online registration; tenants have a legal right to install; standard Schuko plug allowed since December 2025	[S18]
Austria, Netherlands, France, Belgium	Legal, 800 W; light or no registration	[S18]
United Kingdom	Legalised from 27 August 2026 (up to 800 W AC)	[S18]
Some European countries	Reported as not generally legal in Hungary and Sweden; for other Central and Eastern European countries we could not confirm (check the grid operator)	[S18]
USA	State by state; Utah first (March 2025, up to 1,200 W); ten states have laws, California signed 30 September 2026	[S18]
Latin America	No plug-in rule found. Any grid-connected inverter needs distributor approval (for example Brazil’s micro-generation rules up to 75 kW, Mexico’s distributed generation under 0.7 MW)	[S31]
Africa	No plug-in rule found. South Africa requires registration of every grid-tied system, even without export	[S32]

Other options when plug-in is not allowed or there is no balcony [Judgement unless stated]:

- **Shared or remote generation:** Brazil allows remote self-consumption, with extra grid charges [S31]; Colombia's 2025 rules cover collective self-generation and energy communities below 5 MW [S31]; Chile has energy-community rules [S31]. In Europe, building-level solar sold to tenants on the same site (e.g. Romania's 2026 prosumer law covers consumers connected to the plant's busbars [S19]).
- **Pay-as-you-go solar kits (Africa and other off-grid areas):** 10.2 million solar energy kits were sold worldwide in 2025 and more than 148 million people use them [S37]. Kits can move with the tenant, need no grid approval, and are paid in instalments by mobile money [S37]. A study of 1,000 households found 77% used less energy after the first year, so systems are often oversized [S37].
- Community energy or a green tariff, where available.
- Off-grid portable power station with foldable panels: legal almost everywhere, but costly per kWh and mostly for backup.
- Efficiency steps that need no permission: heat-pump or solar hot water, induction hob or electric pressure cooker, smart plugs, tariff switching.

Suggested rule for the app

Ask "own / rent / flat in a shared building" first. For renters, show only options that need no landlord works, and mark each with its legal status for the user's country. Where the status is unknown, show "check with the grid operator and landlord".

7. Upkeep and hands-on work

Option	Upkeep (hours per year; technician)	Notes
Rooftop PV	About 1–2 h/yr (look, occasional rinse) [Judgement]; technician every 5 years [S26]	String inverter replaced once in the life of the panels (10–15 yr) [S26].
Rooftop PV in dusty or rain-free zones	Rinse every 2–4 weeks in dry season: about 10–25 h/yr [Judgement]. Plant studies in the Sahel and deserts find best cleaning cycles of about 14–20 days [S36].	Cleaning needs water; use soft water and early morning.
Battery	Almost none; monitoring	Replacement planning after 10–15 years [S26].
Heat pump (air or ground)	1–2 h/yr (filters, clear coil) [Judgement]; **annual professional inspection** recommended [S26]	Ground loops need no service once in.
Solar water heater	About 1 h/yr; check every 1–2 years [Judgement]	Stagnation in summer must be designed out [S16]; hard water scale in some regions [Judgement].
Small wind	Annual inspection; budget about **1% of installed cost per year** for O&M [S17]	Moving parts wear faster than PV.
Micro-hydro	**High**: clear the screen of debris often, especially in the wet season and after floods [Judgement]	Also water-rights compliance [S21].
Pellet boiler	Sweep the boiler **weekly** [S23]; ash removal; ordering pellets; yearly service	Needs a fuel store [S23].
Log (gasification) boiler	Manual loading: a fill burns up to about 4 h; typically daily in winter [S23]; plus wood handling (several hours/week) [Judgement]	Needs a 300 L+ buffer tank [S23].
Biogas digester	**Daily feeding** [S22]; about 20–40 min/day [Judgement]	Not for people who do not keep livestock.

Suggested rule for the app

"No upkeep" profile: recommend **rooftop PV (optionally with battery) with a monitoring or service plan, and an air-source heat pump with a service contract** where heating is needed. Where possible mention leasing, pay-as-you-go or a third-party-owned system. Exclude wind, hydro, biomass and biogas for this profile.

8. What matters most: profile weights

- Not answered, I need all five set of weights to determine if they look right.

9. Questions based on the maps

River (small hydro)

Power from a small hydro site is about **5 watts per metre of head per litre per second** of flow [Derived from the US Department of Energy formula: head in feet × flow in gallons per minute ÷ 10 [S21]]. Head below about 3 m (10 ft) is generally uneconomical for micro-hydro, and about 0.6 L/s (10 gallons per minute) is the lower limit of flow; a trade-off between head and flow applies [S21]. Pipe (penstock) is often 30–50% of the cost [S21], and the head must be real vertical drop, not distance [S21]. Mountain areas (Andes, Central America, East African highlands) have many such sites [Judgement].

Suggested rule for the app

Mention hydro only if all hold: a stream within **500 m**, at least **3 m of drop** capturable in a pipe of up to about 300 m, and **at least 10 L/s all year** (check the dry season, which is long in many tropical and Mediterranean zones). Expected power = 5 × head (m) × flow (L/s) W; show hydro only if at least about 300 W continuous [Judgement].

Example: 5 m and 20 L/s gives about 500 W, or roughly 4,400 kWh/yr running all year [Derived].

Add a warning on water rights and minimum river flow [S21][S27].

Wind

3 m/s at 20 m is too low. Grid-connected small wind is practical from about 4 m/s annual average [S17], and 4.5–5.5 m/s at hub height is considered the minimum for cost-effective systems [S17]. Small turbines have an average capacity factor of only 17% (range 2–36%) [S17].

Terrain around the turbine	Wind speed at 20 m vs open farmland	Power vs open farmland
Open farmland	100%	100%
A few trees	96%	88%
Many trees, few buildings	89%	71%
Forest / woodland	72%	37%
Suburb	65%	28%
City centre	under 20%	about 0%

Table is [Derived]: log wind profile with surface roughness lengths of 0.03, 0.1, 0.25, 0.5, 1.5 and 3 m [S17] and assumed displacement heights of 0, 0, 2, 10, 7 and 15 m, same wind at 100 m. Raising the hub from 20 to 30 m adds only 6% (open) to 14% (city) [Derived; wind-shear exponents 0.14–0.33 in [S17]].

Suggested rule for the app

Show wind as "possible" at **≥4.5 m/s at hub height** and "good" at **≥5.5 m/s**. Convert map speeds to hub height and multiply by the terrain factor above.

Add a siting rule: blades at least 10 m above any obstacle within 150 m; obstacles disturb air to twice their height and up to 20× their height downwind [S17]. Do not recommend roof-mounted wind in built-up areas.

Animals and food waste (biogas)

Worth mentioning when at least **20–40 kg of manure per day** can be collected: **3–5 head of stabled cattle, 8–12 pigs or 16–20 sheep or goats**, enough for cooking for a family of 4–6 [S22]. One cow gives about 9–15 kg dung and 0.4 m³ gas per day; cooking needs about 0.3–0.9 m³ per person per day [S22]. **Food waste alone is not enough**: about 20–30 L of biogas per person per day against 300–900 L needed for cooking [S22]. Digesters work best at 25–35 °C [S22], so **tropical and warm-highland zones are the best fit**; in cold zones they need heating, so flag them only for larger herds [Judgement].

Biomass and firewood

- **Log gasification boiler:** good where the user has cheap local wood, a cellar or shed, and accepts manual loading. Needs a buffer tank of about 50 L per kW (at least 300 L) and loading about every 4 h of burn [S23].
- **Pellet boiler:** good for a high heat demand, radiator systems and homes where a heat pump cannot meet the load, with space for a fuel store; pellets carry about 5 MWh per tonne against 2–3 for logs [S23].
- **Not recommended** in dense areas with air-quality problems [S23], or for low-demand, well-insulated homes where a heat pump is simpler [Judgement]. A sensible minimum is a rural or edge-of-village home with heat demand above about 15 MWh/yr [Judgement].
- **Hot-humid zones and Africa:** the need is cooking, not heating. The baseline is firewood or charcoal for most households [S34], so recommend **clean cooking** first: improved stoves, LPG, or an electric pressure cooker where electricity is reliable [S34].
- In Central and Eastern Europe firewood is often already the baseline heat source, so a boiler recommendation should be compared with it [S27].

10. Areas to keep free

Where	Rule	Source
UK (domestic roof edge)	Modules not within 400 mm of any roof edge unless wind-uplift measures are shown	MIS 3002 [S24]
USA, homes	457 mm (18 in) from ridge if panels cover up to 33% of roof; 914 mm (36 in) above that; two 914 mm pathways edge to ridge	IRC [S24]
USA, larger buildings	1.8 m (6 ft) perimeter, or 1.2 m (4 ft) if either side is 76 m or less	IFC [S24]
Italy	1 m clear strip at roof edge; 1 m around vents; groups of panels 20 m or less with 2 m paths	Fire service guideline 2025 [S24]
Netherlands	Clear edge at least as wide as the highest panel point; 1 m around rooflights	Fire service advice [S24]
France (public buildings)	0.90 m perimeter and paths; fields of 300 m² or less	CCS opinion; local fire services differ [S24]
Germany	Insurer guides: 2.5 m from a fire wall; inverter 0.5 m from combustible material	VdS [S24]
India	75 cm roof access paths	CEA 2023 [S24]
Latin America, Africa	No national roof-setback rule found in our research	Use the defaults below and local fire guidance [Judgement]

Suggested rule for the app

Defaults: **0.5 m from roof edges and ridge** (covers the UK 400 mm and US 457 mm rules), **1 m around roof windows, vents and fire walls**, and a ridge/pathway of 0.9 m on larger arrays. Use the country rule where the app knows it, and say “confirm with local fire service” elsewhere. In hurricane and strong-wind areas (Caribbean, coasts) keep to the wider edge strips and use certified fixings [Judgement].

Chimneys: keep out of the chimney’s shadow (check with a shading simulation) and leave at least 1 m for sweeping access [Judgement].

Plot boundary: no universal value. For small wind use about **1.5× tower height** [S17]; for ground PV and boreholes use local planning rules (boreholes 5–10 m apart) and show “check local rules” [S15].

11. Costs and lifetimes

Cost bands

Prices differ by a factor of five between regions, so one set of euro bands cannot work worldwide. We suggest **global bands in US dollars per watt, with regional examples**, plus a user-typed local quote. Installed prices for a typical home system, before incentives:

Option	Global bands (installed)	Regional examples	Source
Rooftop PV (grid-tied)	Low: under $1.0/W · Medium: $1.0–2.0/W · High: over $2.0/W	Central and Eastern Europe about €800–1,200/kWp (5 kWp: €4,000–6,000); Western Europe €1,200–1,800/kWp; Brazil R$2.45/Wp on average in early 2026 (a 5 kWp system R$13,000–25,000, about $0.4–0.5/W) [exchange rate Judgement]; Australia about $0.6–0.9/W; India ₹40–55/Wp; South Africa grid-tied 5 kW about R85,000–120,000; USA $2.3–3.6/W	[S25][S33]
Hybrid system with battery (solar + inverter + battery)	Low: under $1,000/kW · Medium: $1,000–1,700/kW · High: over $1,700/kW (of PV size, battery included) [Judgement]	South Africa: 5 kW inverter with 5–10 kWh battery about R75,000–145,000 (all in); Nigeria: 6 kWp + 15 kWh + 5 kW inverter about $6,200–9,400; generic Africa: 5 kW system $3,500–5,000 (vendor estimate)	[S33]
Home battery (installed)	Low: under $300/kWh · Medium: $300–600 · High: over $600	South Africa about R8,500/kWh installed; Germany quotes range from €270/kWh to over €1,000/kWh for a full system. **Sources disagree widely; refresh with local quotes.**	[S25][S33]
Air-to-water heat pump (cold and cool zones)	Low: under $8,000 · Medium: $8,000–15,000 · High: over $15,000	Central and Eastern Europe €6,000–18,000 per house; Netherlands €8,000–18,000. Ground-source adds €5,000–15,000 for boreholes.	[S25]
Solar water heater	Not verified; ask installers [Judgement: about $1,500–4,000 for 4–6 m² with tank]	—	—
Small wind	Not verified; one 1.2 kW rooftop example cost about $11,000 installed	—	[S17]
Micro-hydro	Site specific; pipe 30–50% of cost; treat as “high” unless a weir exists	—	[S21]
Pellet / log boiler	Not verified outside the UK: log boiler about £5,000, pellet boiler about £15,000	—	[S23]
Biogas digester, clean cookstove	Not researched	—	—

Exchange rates for the regional examples change; keep prices in local currency and convert at run time. Brazil’s price level moves with import duties; a 2026 study reports a 7% rise in the first half of 2026 [S33].

Lifetimes to use

Component	Use (years)	Range found	Source
Solar panels	25	25–30; output falls about 0.5–0.8% a year	[S26]
String inverter	12	10–15 (microinverters 20–25)	[S26]

Component	Use (years)	Range found	Source
Home battery (LFP)	12	10–15 warranty and life; 6,000–10,000 cycles; some sources say 15–20. Shorter in hot, dusty sites [Judgement]	[S26]
Heat pump (air-to-water)	15	15–20; 12–18 elsewhere	[S26]
Small hydro	20	“20 or more years” for run-of-river systems	[S21]
Ground loop, solar thermal, small wind, boilers	50 / 20 / 20 / 15	Not verified	[Judgement]

12. Rules and permits

Rules change often and differ by city. The app should show them as “typical requirements, confirm locally”, with a review date. Where we found nothing, the table says so.

Option	Europe (incl. Central and Eastern)	North America	Latin America	Africa
Rooftop PV and export	Grid-operator notification and a compliant inverter are usual. Net metering is closing in many countries: in Hungary new household systems have gross settlement with grid-communicating inverters and about HUF 5/kWh export (2024) [S20]; Romania’s 2026 prosumer law has size tiers (≤27, 27–200, 200–400 kW) and time-limited compensation [S19].	USA: state-by-state net metering [S18]. Canada: utility net metering [S31].	Brazil: micro-generation ≤75 kW, 60-month credits, grid fee (Fio B) at 60% in 2026; pre-2023 filings keep old rules to 2045 [S31]. Mexico: distributed generation under 0.7 MW is permit-exempt (2025 law) [S31]. Chile: net billing ≤300 kW, surplus paid at node price, registered installer [S31]. Argentina: user-generator, small ≤3 kW [S31]. Colombia: net metering and collective self-generation [S31].	South Africa: register every grid-tied system (≤100 kW with Eskom or the municipality), NRS 097-2-1:2024 inverter, compliance certificate; net billing approved December 2024; fee waiver for ≤50 kVA ran to 30 Sept 2026, so check for an extension [S32]. Kenya 2024 net-metering rules; Ghana rollout; Egypt net metering; Rwanda no formal net metering; Morocco residential low-voltage framework pending as of April 2026 [S32].
Small wind	Municipal construction approval; Sweden building permit [S27]. Hungary’s old 12 km rule was repealed but grid connection limits new wind [S27].	Local zoning; height, setback (about 1.5× tower) and noise limits are common [S17].	Not researched; expect municipal and environmental approval [Judgement].	Not researched [Judgement].
Small hydro	Water-rights or water-management permit and environmental approval (e.g. Romania Water Law 107/1996; Hungary water rights) [S27].	State water rights and minimum in-stream flows [S21].	Not researched; a water-use concession and environmental licence are normal [Judgement].	Not researched [Judgement].
Ground-source heat pump boreholes	Hungary: deeper than 20 m needs a building permit and, since March 2023, a Mining Authority licence for geothermal use; UK: normally no planning permission; Sweden: notify or apply [S27]. Other countries not confirmed.	Local and state rules [Judgement].	Rarely relevant (not researched).	Rarely relevant (not researched).

Option	Europe (incl. Central and Eastern)	North America	Latin America	Africa
Air heat pump, solar water heater, battery	Generally no special permit; installer certification and refrigerant rules apply [Judgement].	Same [Judgement].	Same [Judgement].	South Africa: hybrid and battery installs need a certificate of compliance [S32].
Biogas / biomass	Special permit regimes in some countries (Hungary) [S27]; air-quality rules for solid-fuel boilers [S23].	Local air rules [Judgement].	Not researched.	Not researched; clean-cooking programmes exist [S34].

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Limits of this research. Several sources are commercial or news pages (installer price guides, vendor blogs). Latin American and African coverage is thinner than European: wind, hydro, ground-source and biogas permits there were not researched, and exchange-rate conversions are approximate. Regulatory statements reflect press and law-firm summaries as of November 2026 and must be confirmed with each national regulator before the app states them as fact.